



Strategic Architectural Roadmap

National Cyber Threat Intelligence Platform — Next-Generation Enhancements

TARGET: AUTONOMOUS CYBER COMMAND CENTER

ENTERPRISE SECURITY & SCALE

1. Visionary Architectural Modules



1. AI Autonomous Active Remediation (Auto-SOAR)

Upgrade playbooks from passive notifications to active eBPF XDP socket drops and iptables firewall rules generated dynamically by Kimi 3 & Claude 3.5 Sonnet ensembles.



2. Neo4j Threat Actor Graph Database

Build a graph database (Neo4j / Memgraph) mapping APT actors → campaign aliases → malware families → infrastructure, rendered via interactive Cytoscape 3D node graphs.



3. FIDO2 / WebAuthn & Vault Secrets

Implement native YubiKey FIDO2 hardware passkey authentication on the NCTIP portal and HashiCorp Vault dynamic zero-trust API key rotation.



4. eBPF Kernel Probes & Sigma Rule Engine

Deploy a C/Go eBPF sensor monitoring kernel syscalls (execve, connect) and a Sigma YAML compiler converting community rules to Elasticsearch DSL queries.

2. High-Availability & Enterprise Scale Upgrades

Architecture Area	Current Implementation	Proposed Enterprise Upgrade	Key Benefit
Database Cluster	PostgreSQL 16 (Single Instance)	CockroachDB / PostgreSQL Patroni Active-Active Cluster	Zero downtime & multi-region failover
Message Streaming	Single Kafka Broker	Kafka MirrorMaker 2 Multi-DC Replication	Regional data sovereignty & disaster recovery
Search Engine	Single Elasticsearch Node	Elasticsearch 3-Node Cluster with Index Lifecycle (ILM)	Long-term multi-terabyte IOC retention
Cyber Deception	Manual Threat Feeds	Integrated Honeypot Network (Cowrie / Dionaea) & Canary Tokens	Real-time perimeter brute-force capture

3. Source Control & System Reference

GitHub Repository: <https://github.com/MuhimAkhtar/CyberThreatIntelligence>

Interactive Swagger API Docs: <http://localhost:3000/api/docs>

Live Development Frontend UI: <http://localhost:5173/>